Healthy Skepticism · Session 4 take-home summary

Fault 6: the syndrome trap

A disease, in the textbook sense, is one thing. There is a mechanism, and because there is a mechanism there is a reasonably predictable course and a treatment aimed at the cause. A syndrome is different. A syndrome is a name for a cluster of findings that tend to travel together. A committee assigns the name, criteria decide the membership, and the criteria are checked off a list.

Naming a cluster is useful. It is how medicine begins to study something it does not yet understand, and some syndromes later graduate to diseases once a mechanism is found. The fault comes when the label starts being treated as if that had already happened: a single condition you have or don't have, with a prognosis of its own and a treatment of its own.

The question that exposes it: *if two people have the same diagnosis, what do they actually have in common?* For a disease the answer is the mechanism. For a syndrome the answer can be close to nothing, and you can establish that by counting.

The specimen: metabolic syndrome

You have metabolic syndrome if you meet any three of these five criteria: waist circumference (40 inches or more for men, 35 for women), triglycerides (150 mg/dL or more), HDL cholesterol (under 40 mg/dL for men, under 50 for women), blood pressure (130/85 mm Hg or more), and fasting glucose (100 mg/dL or more), in each case counting anyone on treatment for the item.

Notice what the checklist is made of. Every criterion is a line drawn across a continuous measurement, so Fault 3, arbitrary categories, is at work five times before the syndrome is even assembled. Today's question is what happens after the lines are drawn.

Criteria: Grundy SM et al. Diagnosis and management of the metabolic syndrome: AHA/NHLBI scientific statement. Circulation 2005;112:2735–2752.

Counting the patient types

Any three of five. Choosing three criteria from five can be done exactly ten ways, and in class we laid out all ten, so the count is checked, not asserted. Each of the ten is a different patient type: a different set of findings, and a different chart. Two people who each qualify on three criteria must share at least one, and one is all they need to share. A weight-and-lipid patient and a pressure-and-sugar patient can hold the same diagnosis with a single overlapping measurement.

The types need different medicine. Read each qualifying combination as a chart and map it to treatment: waist calls for weight management, the two lipid criteria for lipid therapy, pressure for antihypertensives, glucose for glucose control. The treatment mix changes from type to type. The syndrome identifies ten different patient types and gives them all one name.

Choosing four criteria adds five more ways in, and taking all five adds one, sixteen qualifying profiles in all, but the ten already make the point. And severity is not on the checklist at all: a person one point over three lines and a person far over all five carry the same label.

The counting follows from the any-three-of-five rule; the treatment mapping follows the AHA/NHLBI direction to treat the components.

No pill for the label

Nearly four decades after the syndrome was named, no drug carries metabolic syndrome as an approved indication. The statement that defines it directs clinicians to treat the components, and in 2005 the American Diabetes Association and its European counterpart published a joint appraisal of the label they had helped build: the criteria are ambiguous, the label predicts little beyond its parts, and clinicians should treat the parts.

Treating the checklist failed when it was tried. Low HDL is one of the five criteria, and for years the number itself became a drug target. Torcetrapib raised HDL by roughly 70 percent and raised deaths; the trial was stopped early. Niacin, added to statins, raised HDL as intended and produced no clinical benefit. Moving a criterion is not treating a disease.

The treatment that finally arrived came by accident. Semaglutide (Ozempic, Wegovy) and the GLP-1 class improve every criterion on the checklist. They were not designed to. They were diabetes drugs, and their wider benefits surfaced because a 2008 FDA rule required cardiovascular safety trials of every new diabetes drug: LEADER, run to show liraglutide did not cause heart attacks, found it prevented them, and SELECT then showed semaglutide cutting cardiovascular events in people with obesity and no diabetes. The drug works because it acts upstream, on the dysregulated energy balance behind the adiposity, a mechanism the checklist never named.

Kahn R et al., Diabetes Care 2005;28:2289–2304. Barter PJ et al., NEJM 2007;357:2109–2122. AIM-HIGH Investigators, NEJM 2011;365:2255–2267. FDA guidance, December 2008. Marso SP et al. (LEADER), NEJM 2016;375:311–322. Lincoff AM et al. (SELECT), NEJM 2023;389:2221–2232.

Spectra stretch the same trap: autism

A spectrum is a syndrome with the variation admitted. In 2013 the DSM-5 folded what had been several diagnoses, including autistic disorder and Asperger's disorder, into a single autism spectrum disorder. One label now spans a non-verbal eight-year-old who needs support around the clock and a chief executive who told a television audience he has Asperger's. Whatever those two have, it is not one disease, and no sentence of the form “autism is caused by X” or “Y helps autism” can be about both of them at once.

The field's own vocabulary has been retreating from the single-disease picture: severity is scored separately on two symptom domains, research finds no single explanation at any level, and practitioners increasingly describe a multi-dimensional spectrum. A cluster of correlated dimensions with a committee-drawn boundary is exactly what a syndrome is. The new vocabulary has walked back to the old concept.

DSM-5, APA 2013. E. Musk, Saturday Night Live, NBC, May 8, 2021. Happé F, Ronald A, Plomin R. Nat Neurosci 2006;9:1218–1220.

Dementia: one umbrella, several diseases

Dementia is not a disease; it is an umbrella over distinct diseases with different mechanisms: Alzheimer's disease (roughly 60 to 70 percent of cases), vascular dementia, Lewy body dementia, frontotemporal dementia, and in community autopsy series the most common finding of all is mixed pathology, more than one at once.

This is the case where treating at the label level can actively harm. Cholinesterase inhibitors are approved for Alzheimer's and nothing is approved for vascular dementia. Antipsychotics prescribed at the label level are hazardous in Lewy body dementia, where severe neuroleptic sensitivity is part of the disease. And the new anti-amyloid drugs work only in early Alzheimer's with confirmed amyloid, which is the GLP-1 pattern again: progress arrives when medicine targets a mechanism inside the label, never the label itself. “A cure for dementia” is a category error; before believing a claim about the umbrella, ask which disease under it the claim could be about.

WHO dementia fact sheet. Schneider JA et al., Neurology 2007;69:2197–2204. O'Brien JT, Thomas A. Lancet 2015;386:1698–1706. McKeith IG et al., Neurology 2017;89:88–100. van Dyck CH et al. (CLARITY AD), NEJM 2023;388:9–21.

ADHD, briefly

The purest specimen sits in the backup slides. ADHD is a checklist of checklists: six or more of nine inattention symptoms, or six or more of nine hyperactivity symptoms. A purely inattentive presentation and a purely hyperactive one can share not a single symptom and carry the same diagnosis; metabolic syndrome at least guarantees its members one criterion in common. And treatment response settles nothing, because stimulants sharpen attention in nearly everyone, shown in healthy children at NIH in 1978. A drug that works on a trait says nothing about whether a label carves out a disease.

DSM-5, APA 2013. Rapoport JL et al., Science 1978;199:560–563.

What to ask

When a syndrome or spectrum turns up in your chart or in a headline, three questions do most of the work. **Which components do I actually have, and how far past each line am I?** The label says three-of-five; your chart says which three, and by how much. **What is known about people with my components, as opposed to people with my label? And is the proposed treatment aimed at a component, a mechanism, or just the name?** Only the first two have evidence behind them.

Ask what the people under a label share. Sometimes the answer is a mechanism, and you are looking at a disease. Often the answer is a checklist.

Next session

The final fault, the causal leap: does bird watching prevent dementia? You now know what “dementia” is, an umbrella over a crowd, and next session you will watch a causal claim get made about that crowd, and learn what it actually took to prove that smoking causes cancer.